

CO2-ASSESSMENT TOOL-2

SCENARIOBASEDASSESSMENT

Name:V.Reddy Sreya

Reg.No:192525124

CourseCode:DSA0103

1. A hospital wants to estimate consultation charges for different patient categories. Design a C++ program for a Consultation Charge System. Design a C++ program for hospital management.

CODE:

```
1 #include <iostream>
2 using namespace std;
3 // Function Prototypes
4 double calculateCharge(double charge);
5 double calculateCharge(double charge, double specialistFee);
6 void updateBill(double &bill, double discount);
7 double addServiceCharge(double bill, double service = 100);
8 int main()
9 {
10     int type;
11     double bill, discount;
12     cout << "1. General\n2. Specialist\nEnter Patient Type: ";
13     cin >> type;
14     if(type == 1)
15         bill = calculateCharge(500);
16     else
17         bill = calculateCharge(500, 300);
18     bill = addServiceCharge(bill);
19     cout << "Enter Discount: ";
20     cin >> discount;
```

```

21     updateBill(bill, discount);
22     if(type == 1)
23         cout << "Patient Type: General\n";
24     else
25         cout << "Patient Type: Specialist\n";
26
27     cout << "Total Consultation Charge = " << bill;
28
29     return 0;
30 }
31 double calculateCharge(double charge)
32 {
33     return charge;
34 }
35 double calculateCharge(double charge, double specialistFee)
36 {
37     return charge + specialistFee;
38 }
39 void updateBill(double &bill, double discount)
40 {

```

```

40 {
41     bill = bill - discount;
42 }
43
44 double addServiceCharge(double bill, double service)
45 {
46     return bill + service;
47 }

```

OUTPUT

```

1. General
2. Specialist
Enter Patient Type: Enter Discount: Patient Type: Specialist
Total Consultation Charge = 900

```

2. Design a C++ program FitnessCentreMembershipSystem. A gym wants to manage member subscriptions, track plans, and calculate membership fees.

CODE:

```
21     else if(plan=="VIP")
22         fee=3000;
23     else
24     {
25         cout<<"Invalid Plan\n";
26         plan="Basic";
27         fee=1000;
28     }
29     totalMembers++;
30 }
31 void display()
32 {
33     cout<<"\nID: "<<id;
34     cout<<"\nName: "<<name;
35     cout<<"\nPlan: "<<plan;
36     cout<<"\nFee: "<<fee<<endl;
37 }
38 static void showTotal()
39 {
40     cout<<"\nTotal Active Members = "<<totalMembers;
```

```
1 #include <iostream>
2 using namespace std;
3 class Member
4 {
5 private:
6     int id;
7     string name, plan;
8     double fee;
9 public:
10     static int totalMembers;
11     void getData()
12     {
13         cin >> id;
14         cin >> name;
15         cout << "Enter Plan (Basic/Premium/VIP): ";
16         cin >> plan;
17         if(plan=="Basic")
18             fee=1000;
19         else if(plan=="Premium")
20             fee=2000;
```

```

40         cout<<"\nTotal Active Members = "<<totalMembers;
41     }
42 };
43 int Member::totalMembers=0;
44 int main()
45 {
46     int n;
47     cout<<"Enter Number of Members: ";
48     cin>>n;
49     Member m[10];
50     for(int i=0;i<n;i++)
51         m[i].getData();
52     cout<<"\nMembership Details\n";
53     for(int i=0;i<n;i++)
54         m[i].display();
55     Member::showTotal();
56     cout<<"\n\nPricing plans can be modified easily by changing the fee values inside
57     return 0;
58 }

```

INPUT:

INPUT

2
101
ravi
premium
102
priya
VIP

OUTPUT:

```

ID: 101
Name: ravi
Plan: Basic
Fee: 1000

ID: 102
Name: priya
Plan: VIP
Fee: 3000

```

3. A bank wants to calculate loan repayment details for customers. Design a C++ program for a Loan Repayment System.

CODE:

PROGRAM EDITOR

C++

Run

Save

INPUT

```
1 #include <iostream>
2 using namespace std;
3 // Function Prototypes
4 inline double simpleInterest(double p, double r, double t);
5 double calculateEMI(double loan, double rate = 8);
6 void updateAmount(double &amount, double subsidy);
7 int main()
8 {
9     double loan, time, interest, payable, subsidy;
10    cin >> loan;
11    cin >> time;
12    interest = simpleInterest(loan, 8, time);
13    payable = loan + interest;
14    cin >> subsidy;
15    updateAmount(payable, subsidy);
16    cout << "\nLoan Amount = " << loan;
17    cout << "\nInterest = " << interest;
18    cout << "\nTotal Repayment = " << payable;
19    return 0;
20 }
```

100000
2
5000

```
21 inline double simpleInterest(double p, double r, double t)
22 {
23     return (p * r * t) / 100;
24 }
25 double calculateEMI(double loan, double rate)
26 {
27     return loan + (loan * rate / 100);
28 }
29 void updateAmount(double &amount, double subsidy)
30 {
31     amount = amount - subsidy;
32 }
```

OUTPUT

```
Loan Amount = 100000
Interest = 16000
Total Repayment = 111000
```

4. A delivery company wants to automate the calculation of parcel delivery charges based on parcel weight, delivery distance, and delivery type. Design and implement a C++ program using functions and object-oriented programming concepts to develop a Parcel Charge Management System.

CODE:

```
1 #include <iostream>
2 using namespace std;
3 class Parcel
4 {
5 private:
6     string type;
7     double weight,distance,packCharge;
8 public:
9     static int totalParcels;
10    void getData(double pack=50)
11    {
12        cin>>type;
13        cin>>weight;
14        cin>>distance;
15        packCharge=pack;
16        totalParcels++;
17    }
18    inline double baseCharge()
19    {
20        return weight*10 + distance*2;
```

```
21    }
22    double calculateCharge()
23    {
24        return baseCharge()+packCharge;
25    }
26    double calculateCharge(double express)
27    {
28        return baseCharge()+packCharge+express;
29    }
30    void display()
31    {
32        cout<<"\nParcel Type: "<<type;
33        cout<<"\nWeight: "<<weight;
34        cout<<"\nDistance: "<<distance;
35        cout<<"\nNormal Charge: "<<calculateCharge();
36        cout<<"\nExpress Charge: "<<calculateCharge(100)<<endl;
37    }
38    static void showTotal()
39    {
40        cout<<"\nTotal Parcels = "<<totalParcels;
```

```
41     }
42 };
43 int Parcel::totalParcels=0;
44 int main()
45 {
46     Parcel p;
47     p.getData();
48     p.display();
49     Parcel::showTotal();
50     return 0;
51 }
```

INPUT:

INPUT

box
5
20

OUTPUT:

OUTPUT

Parcel Type: box
Weight: 5
Distance: 20
Normal Charge: 140
Express Charge: 240

Total Parcels = 1

5. A utility company wants to automate the calculation of water bills for domestic and commercial consumers. Design and implement a C++ program using functions and object-oriented programming concepts to develop a Water Billing System. Create a class Consumer with appropriate private data members such as consumer ID, consumer type, water consumption (in liters), and bill amount.

CODE:

```
1 #include <iostream>
2 using namespace std;
3 // Function Prototypes
4 double calculateBill(double liters);
5 double calculateBill(double liters, int commercial);
6 void applySubsidy(double &bill, double subsidy);
7 double addTax(double bill, double tax=5);
8 class Consumer
9 {
10 private:
11     int id;
12     string type;
13     double liters, bill;
14 public:
15     static int totalConsumers;
16     void input()
17     {
18         cin >> id;
19         cin >> type;
20         cin >> liters;
```

```
21         if(type=="Domestic")
22             bill=calculateBill(liters);
23         else
24             bill=calculateBill(liters,1);
25         bill=addTax(bill);
26         double subsidy;
27         cin >> subsidy;
28         applySubsidy(bill, subsidy);
29         totalConsumers++;
30     }
31     void display()
32     {
33         cout << "\nConsumer ID: " << id;
34         cout << "\nType: " << type;
35         cout << "\nWater Consumption: " << liters;
36         cout << "\nFinal Bill: " << bill << endl;
37     }
38     static void showTotal()
39     {
40         cout << "\nTotal Consumers = " << totalConsumers;
```



```

41     }
42 };
43 int Consumer::totalConsumers=0;
44
45 double calculateBill(double liters)
46 {
47     return liters*2;
48 }
49
50 double calculateBill(double liters,int commercial)
51 {
52     return liters*5;
53 }
54 void applySubsidy(double &bill,double subsidy)
55 {
56     bill=bill-subsidy;
57 }
58 double addTax(double bill,double tax)
59 {
60     return bill+(bill*tax/100);

```

```

61 }
62 int main()
63 {
64     int n;
65     cout<<"Enter Number of Consumers: ";
66     cin>>n;
67     Consumer c[10];
68     for(int i=0;i<n;i++)
69         c[i].input();
70     cout<<"\nConsumer Details\n";
71     for(int i=0;i<n;i++)
72         c[i].display();
73     Consumer::showTotal();
74     return 0;
75 }

```

INPUT:

INPUT

2
101
Domestic
1000
100
102
commercial
2000
200

OUTPUT:

OUTPUT

```
Consumer ID: 101  
Type: Domestic  
Water Consumption: 1000  
Final Bill: 2000  
  
Consumer ID: 102  
Type: commercial  
Water Consumption: 2000  
Final Bill: 10300
```